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Deciphering Conformational Changes Associated with the Maturation of Thrombin Anion Binding Exosite I

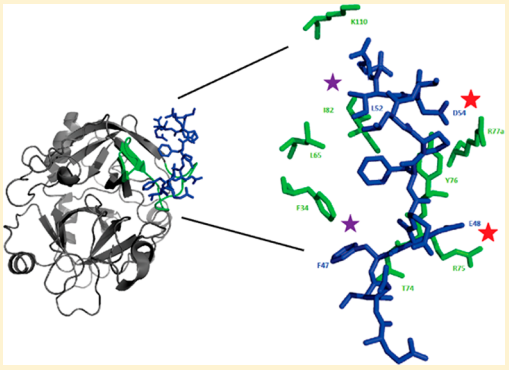
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Supporting Information

ABSTRACT: Thrombin participates in procoagulation, anticoagulation, and platelet activation. This enzyme contains anion binding exosites, ABE I and ABE II, which attract regulatory biomolecules. As prothrombin is activated to thrombin, pro-ABE I is converted into mature ABE I. Unexpectedly, certain ligands can bind to pro-ABE I specifically. Moreover, knowledge of changes in conformation and affinity that occur at the individual residue level as pro-ABE I is converted to ABE I is lacking. Such changes are transient and were not captured by crystallography. Therefore, we employed nuclear magnetic resonance (NMR) titrations to monitor development of ABE I using peptides based on protease-activated receptor 3 (PAR3). Proton line broadening NMR revealed that PAR3 (44−56) and more weakly binding PAR3G (44−56) could already interact with pro-ABE I on prothrombin. $^{1}H-^{15}N$ heteronuclear single-quantum coherence NMR titrations were then used to probe binding of individual ^{15}N-labeled PAR3G residues (F47, E48, L52, and D54). PAR3G E48 and D54 could interact electrostatically with prothrombin and tightened upon thrombin maturation. The higher affinity for PAR3G D54 suggests the region surrounding thrombin R77a is better oriented to bind D54 than the interaction between PAR3G E48 and thrombin R75. Aromatic PAR3G F47 and aliphatic L52 both reported on significant changes in the chemical environment upon conversion of prothrombin to thrombin. The ABE I region surrounding the 30s loop was more affected than the hydrophobic pocket (F34, L65, and I82). Our NMR titrations demonstrate that PAR3 residues document structural rearrangements occurring during exosite maturation that are missed by reported X-ray crystal structures.



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Thrombin (factor IIa) is a multifunctional enzyme that plays critical roles associated with procoagulation and anticoagulation. Procoagulant functions of thrombin include activating other coagulation proteins (zymogen factors V, VIII, and XIII), converting fibrinogen into fibrin that then polymerizes into a blood clot, and helping to activate platelets by cleaving the protease-activated receptors (PARs). An important anticoagulant function of thrombin is to activate protein C and thereby help inhibit a set of coagulant proteins.[1]

Thrombin substrate specificity is regulated by a serine protease active site region, a series of surface loops, and two distant exosites. Similar to those of trypsin and chymotrypsin,[2] the catalytic triad of thrombin is composed of His, Asp, and Ser[3] (Figure 1A). The thrombin 60s (or β-insertion) loop regulates entrance into the active site, and the autolysis (or γ) loop helps further control substrate specificity. In addition, thrombin is allosterically controlled by a Na^{+} binding site.[4,5] Thrombin also has two anion binding exosites, ABE I and ABE II, located on opposite sides of the serine protease active site (Figure 1A). These exosites have been shown to direct substrates to the active site, contribute to opening of the active site region, and help in attracting other regulatory molecules. Ligands that bind to ABE I include fibrinogen,[6] thrombomodulin,[7] protease-activated receptors PAR1[8] and PAR3,[9] and the leech-derived inhibitor hirudin.[10,11] Ligands that target ABE II include heparin,[12] heparin analogues,[13] factor VIII,[14] fibrinogen γ',[15] and platelet receptor protein GpIbα.[16] Direct allosteric linkage may exist between ABE I and ABE II to modulate thrombin activity.[17−20]

Thrombin is expressed as the zymogen prothrombin (ProT) (Figure 1B).[21,22] As ProT is proteolytically converted to thrombin, the immature pro-ABEs develop into the mature, active ABE I and ABE II (Figure 1C).[21,23] In ProT, access to pro-ABE II is blocked by the kringle-containing fragment 1.2 (F1.2). This pro-ABE II region becomes exposed during the cleavage and activation process.[21,24,25] By contrast, the pro-ABE I region is not directly affected by a cleavage event. Instead, subsequent conformational changes likely lead to ABE I maturation.[21,26]

Designing new therapeutic anticoagulants that are directed toward individual thrombin exosites is a promising area of investigation.[13,27,28] Different segments of the mature exosite

Received: September 28, 2017
Revised: October 24, 2017
Published: November 7, 2017

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DOI: 10.1021/acs.biochem.7b00970
Biochemistry 2017, 56, 6343−6354

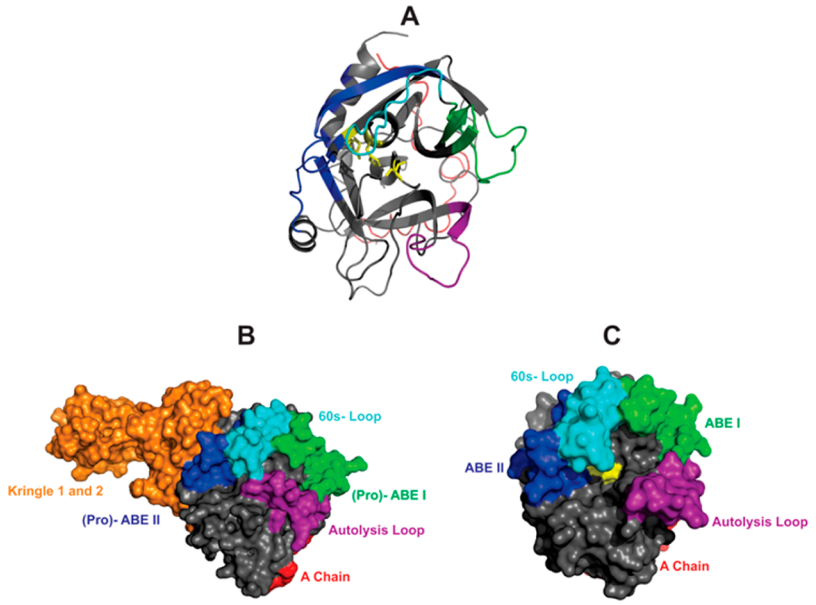


Figure 1. Crystal structures of thrombin [Protein Data Bank (PDB) entry 1PPB] and prothrombin (PDB entry 4HZH). (A) Ribbon diagram of thrombin in which the catalytic triad (His 57, Asp 102, and Ser 195) is colored yellow, (Pro)- ABE I is colored green, (Pro)- ABE II is colored blue, the autolysis loop is colored purple, the 60s loop is colored cyan, and the A chain is colored red. (B) Space filling model of zymogen prothrombin missing the Gla domain. Kringles 1 and 2 are colored orange. The other colored regions match those shown in panel A. (C) Thrombin is generated from prothrombin following cleavages after Arg 271 and Arg 320. The space filling model highlights the removal of the kringles (orange), the exposure of the active site (yellow), and full maturation of ABE I (green) and ABE II (blue).

surfaces could be targeted without directly occluding the active site with a small molecule. Unexpectedly, there are reports that ABE I-directed ligands based on hirudin[29,30] and DNA/RNA aptamers[31,32] can already bind to the immature pro-ABE I on ProT.

Similar to hirudin and the aptamers, PAR3 also contains a binding region directed to thrombin ABE I.[9] PARs are members of the G-protein-coupled receptor family and are involved in platelet activation,[33] vascular remodeling, and vascular permeability.[34] A fragment of PAR3 including amino acid residues ^{44}QNTFEEFPLSDIE^{56} has been confirmed to bind in a specific manner to thrombin ABE I by X-ray crystallography (Figure 2).[9] A review of the crystal structure reveals that thrombin residues located <4 Å from the PAR3 include F34, L65, R75, Y76, R77a, I82, and K110. Chymotrypsin numbering is used to define the thrombin sequence. Thrombin R77a is an extra amino acid residue located after E77, which is not found in chymotrypsin.

The PAR3 (44−56) sequence is similar to the ABE I-directed region found within the leech-derived inhibitor hirudin (Figure 2).[10] PAR3 could thus be a valuable new system for helping to decipher the specific conformational changes that occur as pro-ABE I on prothrombin matures to ABE I on thrombin. An examination of the X-ray crystal structures of prothrombin,[24] thrombin-PAR3,[9] and active site-inhibited PPACK-thrombin[3] reveals no striking differences between their pro-ABE I and ABE I regions (Figure 3).

The backbone root-mean-square deviation[35] values for these three anion binding exosite regions were <0.8 Å. These values indicate that conformational changes may have been missed by crystallographic studies and are transient in nature. Nuclear magnetic resonance (NMR) could thus provide a critical alternative strategy for directly monitoring exosite maturation. To achieve this goal, we have performed NMR titration projects comparing the binding of PAR3 amino acid residues

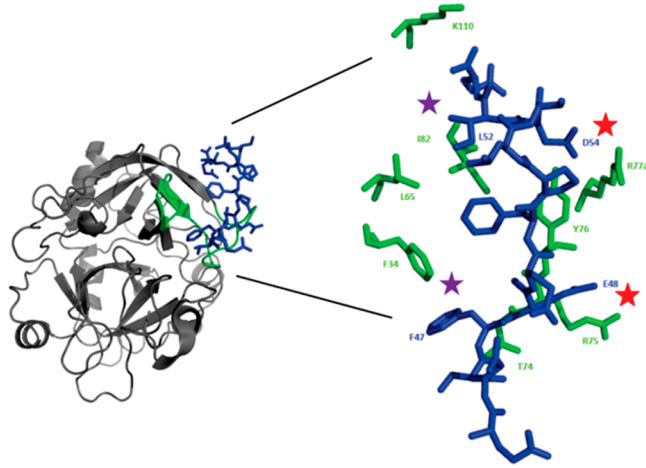


Figure 2. Crystal structure of thrombin in complex with murine PAR3 fragment (44−56). The thrombin surface is rendered as gray ribbons and the PAR3 peptide as blue sticks. Thrombin ABE I residues that are located <4 Å from PAR3 (44−56) are displayed as green ribbons or green sticks. The PAR3 residues later chosen for ^{15}N HSQC titration studies include acidic residues D54 and E48 (red stars) and hydrophobic residues F47 and L52 (purple stars).

(44−56) to the original zymogen state ProT versus the final active protease thrombin. For the first time, the ability of the PAR3 fragment (44−56) to target the immature (pro)-ABE I site on ProT could be explored.

Previous hydrogen−deuterium exchange studies coupled with mass spectrometry (HDX−MS) demonstrated that regions of pro-ABE I experience increases in their levels of solvent exposure as ProT is converted to thrombin.[23] These exposures are hypothesized to be part of the exosite maturation process. NMR removes the issue of only probing segments of a protein and would allow for single-amino acid residue analysis.

DOI: 10.1021/acs.biochem.7b00970
Biochemistry 2017, 56, 6343−6354



Figure 3. Structural alignment of the ABE I regions of prothrombin (PDB entry 4HZH), thrombin-PAR3 (PDB entry 2PUX), and PPACK-thrombin (PDB entry 1PPB). Regions encompassing exosite I (residues F34−M84, thrombin straight numbering) were selected, and then a backbone alignment was performed using MolMol. The backbone ABE I root-mean-square deviation values for the different protein−protein comparisons were then determined: 0.79 Å for ProT and thrombin-PAR3, 0.47 Å for PPACK-thrombin and thrombin-PAR3, and 0.63 Å for ProT and PPACK-thrombin. This analysis reveals no striking differences for the backbone atoms located in ABE I.

Moreover, NMR has the unique advantage of allowing K_D measurements in solution at this individual residue level.

Thrombin is highly dynamic and adopts distinct states in the presence of ligands, which enable this enzyme to fulfill its array of protease activities.[36−40] Previous NMR studies of thrombin revealed that there are key residues within the 30s and 70s loop regions of ABE I that cannot be monitored by NMR.[36−40] Some of these thrombin residues become NMR-visible following introduction of ABE I ligands, whereas others do not. Intriguingly, surface loops and the ABE I region can remain flexible even when the active site of thrombin is inhibited.[37,38,40] Because of the flexible nature of thrombin, development of the mature 30s and 70s loop regions can be challenging to probe. To overcome this issue, a novel NMR titration approach was needed. For the current project, the PAR3 (44−56) peptides were labeled with ^{15}N at the amide nitrogen of specific amino acid residues, and ProT and thrombin remained unlabeled.[41]

Critical NMR information about how the PAR3 peptide (44−56) responds to the changing environment that occurs upon exosite maturation could now be collected. NMR titration studies first revealed that the native PAR3 (44−56) sequence bound too tightly to be monitored by NMR. A P51G substitution was successful in decreasing the PAR3 affinity. One-dimensional (1D) ^{1}H line broadening NMR studies revealed that PAR3G (44−56) could bind to immature pro-ABE I on ProT and also mature ABE I on thrombin. ^{1}H−^{15}N heteronuclear single-quantum coherence (HSQC) NMR titrations were then performed with selectively labeled PAR3G peptides. Acidic PAR3G [^{15}N]E48 and [^{15}N]D54 both entered into higher-affinity, intermediate exchange regimes as ProT was converted to thrombin. Strong binding events involving PAR3G D54 could be weakened with thrombin mutant R77aA, thus supporting the presence of a salt bridge. By contrast, PAR3G [^{15}N]F47 and [^{15}N]L52 both bound more weakly to ProT/thrombin than the acidic PAR3G residues did. Furthermore, [^{15}N]F47 and [^{15}N]L52 were able to document important changes to the hydrophobic exosite environment as pro-ABE I matures to ABE I.

MATERIALS AND METHODS

Materials. Human plasma prothrombin (ProT) and thrombin were purchased from Haematologic Technologies, Inc. (Essex Junction, VT). Recombinant thrombin mutant R77aA (chymotrypsin numbering) was a kind gift from E. Di Cera and L. Pelc (St. Louis University School of Medicine, St. Louis, MO).[42] R77a is an extra thrombin amino acid residue located after E77, which is not found in chymotrypsin. For the R77aA project, basic R77a was replaced with a small, neutral A. The active sites of the thrombin species were protected from autolysis by blocking with PPACK (D-phenylalanyl-L-prolyl-L-arginine chloromethyl ketone). This serine protease active site inhibitor was purchased from Calbiochem (San Diego, CA). D_2O (99.96%) was from Cambridge Isotope Laboratories (Andover, MA).

Synthetic Peptides. A series of PAR3-based peptides were custom synthesized by New England Peptide (Gardner, MA). PAR3 (^{44}Q N T **^{15}F** E E F P L S **^{15}D** I E^{56}), abbreviated as PAR3 (44−56), is the original PAR3 sequence, and amino acid residues F47 and D54 were labeled with ^{15}N at their amide nitrogens. PAR3 (44−56, P51G, ^{15}F47, ^{15}D54) (^{44}Q N T **^{15}F** E E F G L S **^{15}D** I E^{56}), abbreviated as PAR3G$_{FD}$ (44−56), contains a Pro51 to Gly substitution (G), and furthermore, F47 and D54 were labeled with ^{15}N. PAR3 (44−56, P51G, ^{15}E48, ^{15}L52) (^{44}Q N T F **^{15}E** E F G **^{15}L** S D I E^{56}), abbreviated as PAR3G$_{EL}$ (44−56), also contains the P51G substitution (G), but now E48 and L52 are labeled with ^{15}N. The purity of each synthesized peptide was verified by high-performance liquid chromatography and matrix-assisted laser desorption ionization time-of-flight mass spectrometry. Initial stock solutions of peptide were solubilized in deionized water, and the concentration was determined by amino acid analysis (AAA Service Laboratory, Inc., Damascus, OR). Peptides were later diluted into 25 mM H_3PO_4, 150 mM NaCl, and 0.2 mM EDTA (pH 6.5) for NMR studies.

The proton chemical shift values for all the PAR3 peptide amino acid residues were determined using a combination of two-dimensional (2D) TOCSY and 2D-transferred NOESY[43,44] experiments performed on a Varian Inova 700 MHz NMR spectrometer. Standard TOCSY and NOESY pulse sequences were employed. As is typical for trNOESY experiments, the ligand−protein complex contained a 10-fold excess of the PAR3 (44−56) peptide fragment, and the chemical shifts report on the solution environment encountered by the peptide in the presence of the target protein.[44] The amide proton ^{1}H chemical shift assignments derived from the TOCSY and trNOESY experiments were later matched with their respective peaks found in ^{1}H−^{15}N HSQC NMR experiments. Both ^{1}H and ^{15}N values could be followed in the HSQC titrations.

Theoretical Basis for 1D Proton Line Broadening NMR and ^{1}H−^{15}N HSQC NMR. 1D proton line broadening studies were performed to monitor binding of the PAR3/PAR3G (44−56) peptide fragment to prothrombin versus thrombin.[43,45,46] Protein−peptide complexes were prepared with a 10-fold excess of peptide. Peptide protons that undergo interactions experience transient on/off events that cause an increase in the proton transverse relaxation rate and/or cause changes in the observed chemical shift position depending on the time scale of the interaction. The resultant alterations in peptide proton line width/shape reflect the weighted contributions of bound and free populations. This NMR line broadening approach can help

DOI: 10.1021/acs.biochem.7b00970
Biochemistry 2017, 56, 6343−6354

map the peptide residue protons that come into direct contact with the protein surface.

$^{1}H-^{15}N$ HSQC NMR titrations were used to assess whether specific ^{15}N-labeled residues located within a peptide ligand exhibited fast, intermediate, or slow exchange on/off a protein surface.[47] In these titrations, changes in chemical shift are monitored as a function of protein−ligand ratios. For weak binding, the interaction is under the fast exchange regime ($k_{ex} \gg |\Delta\omega|$), where k_{ex} is the exchange rate of the interaction and $\Delta\omega$ is the resonance frequency difference between the bound and free states. Only one signal per ^{15}N-labeled residue is observed, reflecting the population-weighted average of bound and free states in terms of chemical shift, intensity, and line width. Furthermore, NMR chemical shifts can be used to quantitatively evaluate affinities for individual ^{15}N-labeled residues.[48,49] For the moderate affinity, intermediate exchange regime ($k_{ex} \approx |\Delta\omega|$), the ^{15}N peak may disappear during the titration because of extensive line broadening. With an increase in the bound population, the peak will later reappear. Finally, for tight binding, in which the interaction is described by the slow exchange regime ($k_{ex} \ll |\Delta\omega|$), two sets of resonances are observed: one corresponding to the free and the other to the bound state. In this situation, the change in the intensity of the peaks will correspond to the relative population of each state during the course of the titration.

Sample Preparation and Analysis for 1D Proton Line Broadening Experiments. All NMR experiments were performed at pH 6.5 and 25 °C. Under these conditions, the highly exchangeable [^{15}N]amide proton could still be readily observed by NMR and the pH was not far from physiological (7.4). To prevent autolysis, the serine protease inhibitor PPACK was added to thrombin at a ratio of 4:1, and the mixture was incubated at 37 °C for 30 min. Previous NMR relaxation studies have confirmed that the ABE I region remains flexible in PPACK-inhibited IIa and can engage in long-range communication across the enzyme surface.[40] Plasma versions of ProT and PPACK-thrombin were buffer exchanged into NMR buffer [25 mM H_3PO_4, 150 mM NaCl, and 0.2 mM EDTA (pH 6.5)] using a Vivaspin 2 ultrafiltration unit with a 5000 Da molecular weight cutoff (Sartorius, Göttingen, Germany). Protein concentrations were determined using extinction coefficients ($E^{1\%}_{280}$) of 18.3 for thrombin and 13.8 for ProT.

Ligand−protein complexes with ratios of at least 10:1 were then prepared. The complexes included 1 mM PAR3 (44−56) with 74 μM ProT, 1 mM PAR3 (44−56) with 77 μM PPACK-thrombin, 960 μM PAR3G (44−56) with 76 μM ProT, and 1 mM PAR3G (44−56) with 76 μM PPACK-thrombin. The peptide at a minimum concentration of 1 mM was used as a free ligand control. The resultant NMR experiments and all others in this project were performed on a Varian Inova 700 MHz spectrometer with a triple-resonance cold probe and pulsed-field Z-axis gradients run at 25 °C. The ^{1}H NMR spectra were processed using Mnova NMR (Mestrelab Research software).

2D $^{1}H-^{13}C$ HSQC Natural Abundance Experiments. Before proceeding to $^{1}H-^{15}N$ HSQC NMR titrations, we performed $^{1}H-^{13}C$ HSQC natural abundance experiments on both PAR3 (44−56) and PAR3G (44−56) to assess whether the proline to glycine substitution would change the PAR3 peptide conformations. For this project, 1 mM PAR3 and 1 mM PAR3G were prepared in NMR buffer [25 mM H_3PO_4, 150 mM NaCl, and 0.2 mM EDTA (pH 6.5)]. Parameters for the 2D $^{1}H-^{13}C$ HSQC project included 128 transients with 64 complex points in the indirect dimension, sweep widths of 8064.5 and 24635.3 Hz for the direct and indirect dimensions, respectively, and 3000 complex points acquired in the direct dimension. 2D $^{1}H-^{13}C$ HSQC data were processed using NMRPipe and nmrDraw and then further visualized using Sparky. The ability to superimpose the spectra of PAR3 (red) and PAR3G (black) confirmed that replacement of proline with flexible glycine did not change the overall peptide structure (Figure S1)

1D and 2D $^{1}H-^{15}N$ HSQC NMR Titration. ProT and active site-inhibited PPACK-thrombin were exchanged into NMR buffer using Vivaspin 2 ultrafiltration units with a 5000 Da molecular weight cutoff. A series of protein−ligand complexes were then prepared involving PAR fragments PAR3 (44−56), PAR3G$_{FD}$ (44−56), and PAR3G$_{EL}$ (44−56). ProT and PPACK-thrombin exhibit limited solubility at the high concentrations typically used for NMR titrations. As a result, an alternative titration strategy was employed for the current NMR project. The protein concentration typically started in the range of 100−200 μM and was serially diluted while maintaining a constant concentration of ^{15}N-labeled ligand. These conditions were achieved by removing a certain volume of the protein−ligand solution and replacing it with the same volume of the ligand solution. The titrations were thus initiated with a high protein:ligand ratio and ended with low protein:ligand ratios all while maintaining a constant peptide ligand concentration. The NMR titrations were therefore monitoring the ability of ProT or thrombin to bind to the ^{15}N-labeled PAR3 fragment.

For the first PAR3 binding studies, the starting complexes included 50 μM PAR3 (44−56) in either 137 μM ProT or 210 μM PPACK-thrombin. The serial dilutions resulted in ProT:PAR3 ratios that spanned from 3:1 to 0.1:1. PPACK-thrombin:PAR3 ratios spanned from 4:1 to 0.1:1. For the PAR3G$_{FD}$ binding studies, starting complexes included 37.5 μM PAR3G$_{FD}$ (44−56) in 70 μM ProT or in PPACK-thrombin. ProT:PAR3G$_{FD}$ ratios during the titrations then spanned from 2:1 to 0.3:1. PPACK-thrombin:PAR3G$_{FD}$ ratios spanned from 2:1 to 0.05:1.

To evaluate the importance of electrostatic interactions between ^{15}N-labeled D54 of PAR3G$_{FD}$ and R77a of wild-type thrombin, a series of 1D and 2D $^{1}H-^{15}N$ HSQC titrations involving recombinant thrombin mutant R77aA and PAR3G$_{FD}$ were performed. The stock solution of R77aA thrombin was in 20 mM Tris (pH 7.4) and 400 mM NaCl. After PPACK treatment, active site-inhibited R77aA thrombin was buffer exchanged into 25 mM H_3PO_4, 150 mM NaCl, and 0.2 mM EDTA (pH 6.5) using a 3 mL Slide-A-Lyzer Dialysis cassette G2 (Thermo Scientific, Rockford, IL) with a 3500 Da molecular weight cutoff and then concentrated using a 5000 Da molecular weight cutoff Vivaspin 2 ultrafiltration unit. Starting complexes included 50 μM PAR3G$_{FD}$ (44−56) with 150 μM PPACK-thrombin (R77aA). PPACK-thrombin (R77aA):PAR3G$_{FD}$ ratios spanned from 3:1 to 0.1:1.

For the PAR3G$_{EL}$ binding studies, the starting complexes included 50 μM PAR3G$_{EL}$ (44−56) with either 180 μM ProT or 211 μM PPACK-thrombin. The protein:ligand ratios of ProT−PAR3G$_{EL}$ and PPACK-thrombin−PAR3G$_{EL}$ complexes spanned from 4:1 to 0.1:1.

Parameters for the 1D $^{1}H-^{15}N$ HSQC titrations included 512 transients, a sweep width of 7022.5 Hz, and 1242 complex points in the direct dimension with PAR3 and 4096 with the PAR3G complexes. Parameters for the 2D $^{1}H-^{15}N$ HSQC

DOI: 10.1021/acs.biochem.7b00970
Biochemistry 2017, 56, 6343−6354

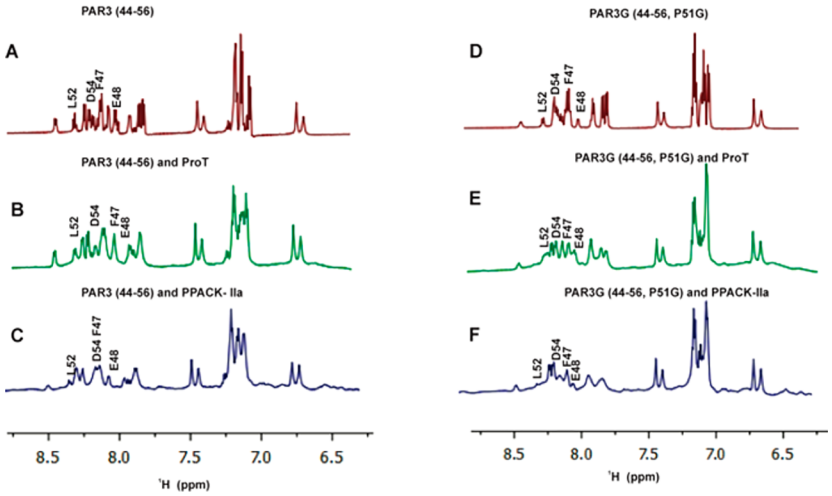


Figure 4. Proton line broadening spectra for PAR3 and PAR3G peptides in the presence of prothrombin and PPACK-thrombin. All NMR samples were in 25 mM H_3PO_4, 150 mM NaCl, 0.2 mM EDTA, and 10% D_2O (pH 6.5). (A) 1D ^{1}H NMR spectrum of 1 mM PAR3 (44−56) in solution. (B) 1D ^{1}H NMR spectrum of 1 mM PAR3 (44−56) in the presence of 74 μM ProT. (C) 1D ^{1}H NMR spectrum of 1 mM PAR3 (44−56) in the presence of 77 μM PPACK-IIa. (D) 1D NMR spectrum of 1 mM PAR3G (44−56) in solution. (E) 1D NMR spectrum of 960 μM PAR3G (44−56) in the presence of 76 μM ProT. (F) 1D NMR spectrum of 1 mM PAR3G (44−56) in the presence of 76 μM PPACK-IIa. Line broadening was observed for residues of PAR3 (44−56) and PAR3G (44−56) when either peptide was bound to prothrombin and PPACK-IIa. The amide protons that were later selected for HSQC titrations are labeled.

titrations included 16 transients with 64 complex points in the indirect dimension, sweep widths of 7022.5 and 1944.3 Hz for the direct and indirect dimensions, respectively, and 1242 complex points acquired in the direct dimension. The ^{15}N-labeled free peptide was used as a control for each titration. 1D ^{1}H−^{15}N HSQC data were stacked using Mnova NMR, and 2D ^{1}H−^{15}N HSQC data were processed using NMRPipe and nmrDraw and then further visualized using Sparky.

Quantitative estimates of binding interactions between individual ^{15}N-labeled peptide ligand residues and specific proteins were determined using in-house scripts written using Python.[50] Information provided to such scripts included the total enzyme ($[P_0]$) and total peptide ($[L_0]$) concentrations employed in the different steps of the HSQC titrations. In addition, the NMR chemical shift difference (Δ_{obs}) between each set of free and bound conditions was provided. For the current NMR project, it is important to note that the peptide ligand concentration was kept constant and the protein was serially diluted. As a result, the NMR titrations were measuring the binding of the protein to a defined peptide ligand concentration. The equation for determining the binding affinity values (K_D) was thus modified so that the denominator now contains $[L_0]$ instead of the more typical $[P_0]$.[47,48]

$$\Delta_{obs} = \Delta_{max}\left[K_D + [L_0] + [P_0] - \sqrt{(K_D + [L_0] + [P_0])^2 - 4[P_0][L_0]}\right]/2[L_0]$$

NMR titrations were performed at least in triplicate to determine K_D values involving plasma-derived ProT and thrombin. Duplicate titrations were performed for thrombin R77aA. An error analysis of the K_D values was performed using a Monte Carlo approach in which a 10% error was imposed on the serially diluted thrombin concentration.[50]

RESULTS

1D Proton Line Broadening NMR Studies with PAR3 (44−56). 1D ^{1}H line broadening NMR experiments were performed on PAR3 (44−56) in the presence of ProT versus PPACK-thrombin (Figure 4A−C). The active site inhibitor PPACK (D-phenylalanyl-L-prolyl-L-arginine chloromethyl ketone) was used to protect thrombin from autolysis during the NMR experiments. Proton (^{1}H) peak broadening could be detected for aliphatic, amide, and aromatic protons of PAR3 in complex with ProT. Such results indicate that a binding surface for PAR3 (44−56) is already available within the immature pro-ABE I on ProT. Substantial peak line broadening was also observed when the peptide was introduced into a solution of PPACK-thrombin containing mature exosite ABE I. Such broadening is impressive considering the lower molecular weight of thrombin (37 kDa) versus that of ProT (72 kDa).[45] The overall, extensive thrombin-induced peak broadening provides further justification that mature ABE I is well suited for accommodating PAR3 (44−56). The ^{1}H chemical shift overlaps observed across the different Figure 4 panels make it difficult to quantitatively evaluate individual proton line effects.

^{15}N HSQC Titration Studies with PAR3 (44−56) Labeled at D54 and F47. An HSQC titration project was designed to systematically characterize binding of PAR3-based peptides that are selectively labeled at amide nitrogen locations. The X-ray crystal structure of the thrombin−PAR3 (44−56) complex was first consulted to select sets of acidic and hydrophobic amino acid residues of PAR3 that display interactions with the 30s and 70s loop regions on thrombin[9] (Figure 2). 2D total correlation spectroscopy (TOCSY) spectra of the bound PAR3 peptide then provided a valuable guide for choosing amino acid residues with amide chemical shift positions that would be less likely to overlap in the proton dimension during the course of the PAR3−protein ^{1}H−^{15}N HSQC titrations.

The first amino acid residues selected included PAR3 F47 and D54. According to the X-ray crystal structure, acidic PAR3 D54 makes a salt bridge contact with thrombin R77a of the 70s ABE I loop whereas PAR3 F47 exhibits $\pi-\pi$ stacking interactions with thrombin F34 in the 30s ABE I loop region[9] (Figure 2). ^{15}N HSQC NMR titrations were thus performed

DOI: 10.1021/acs.biochem.7b00970
Biochemistry 2017, 56, 6343−6354

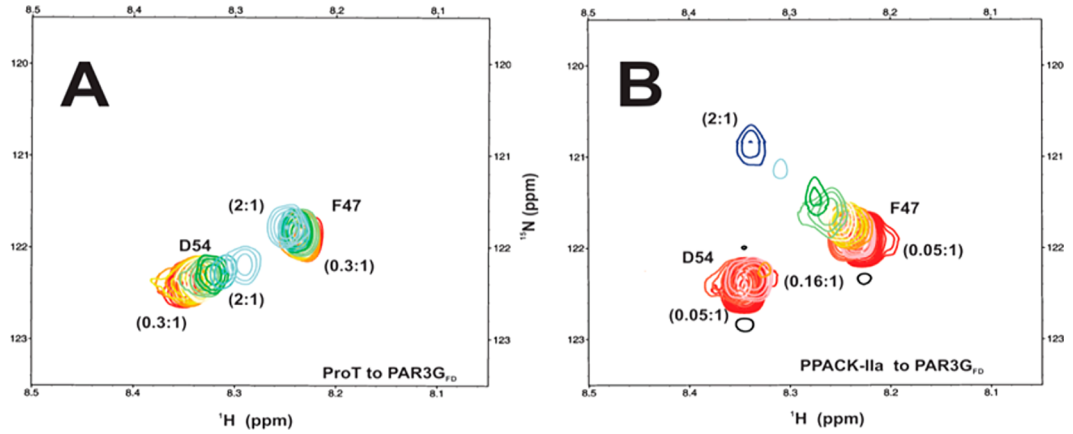


Figure 5. 2D ^{1}H−^{15}N HSQC NMR titrations of PAR3G$_{FD}$ (44−56) in the presence of ProT and PPACK-IIa. All NMR samples were in 25 mM H_3PO_4, 150 mM NaCl, 0.2 mM EDTA, and 10% D_2O (pH 6.5). (A) For the PAR3G$_{FD}$ binding studies with ProT, starting complexes included 37.5 μM PAR3G$_{FD}$ (44−56, [^{15}N]F47, [^{15}N]D54) in 70 μM ProT. The serial dilutions resulted in ProT:PAR3G$_{FD}$ ratios that spanned from 2:1 to 0.3:1. (B) For PPACK-IIa, starting complexes included 37.5 μM PAR3G$_{FD}$ (44−56, [^{15}N]F47, [^{15}N]D54) in 70 μM PPACK-IIa. The serial dilutions resulted in PPACK-IIa:PAR3G$_{FD}$ ratios that spanned from 2:1 to 0.05:1. Representative data sets are shown. Colors for the HSQC crosspeaks span from blue (highest protein:peptide ratio) to red (free peptide).

with PAR3 (44−56) labeled with ^{15}N at D54 and F47. These two PAR3 residues both exhibited proton line broadening in the presence of ProT and PPACK-thrombin, confirming contact with the target protein in solution (Figure 4A−C).

The HSQC titrations started out with a 3:1 ProT:PAR3 ratio (containing the [^{15}N]D54 and [^{15}N]F47 residues). Extensive peak broadening was observed for both [^{15}N]D54 and [^{15}N]F47 until the ProT:PAR3 ratios were serially diluted to 0.3:1 (D54) or 0.6:1 (F47) and lower. With PPACK-thrombin, peaks for both PAR3 residues could not be detected until ratios of ≤0.3:1 were reached (Figures S2 and S3). These NMR results suggest that PAR3 D54 and F47 can already interact with pro-ABE I on ProT and tightens in the presence of PPACK-thrombin with a mature ABE I.

Proton Line Broadening NMR and ^{15}N HSQC NMR Titration Studies with PAR3G$_{FD}$ (44−56) Labeled at D54 and F47. The extensive line broadening observed for PAR3 (44−56) bound to PPACK-IIa was a major hindrance for ^{15}N NMR titration studies, and binding affinities could not be determined. To weaken this interaction, a modified sequence was designed in which the P51 was replaced with a flexible glycine (P51G). The resultant PAR3G$_{FD}$ (44−56) showed 1D proton line broadening with both ProT and PPACK-thrombin (Figure 4D-F). Natural abundance ^{13}C HSQC spectra were recorded for both PAR3 peptides. As seen in Figure S1, the two spectra show good overlap, suggesting that the Pro to Gly substitution has not caused substantial changes to the conformation or the chemical environments of the two peptides.

^{1}H−^{15}N HSQC titrations were then performed with PAR3G$_{FD}$ containing ^{15}N-labeled D54 and F47. The ProT:PAR3G$_{FD}$ ratios spanned from 2:1 to 0.3:1, and PPACK-thrombin:PAR3G$_{FD}$ ratios spanned from 2:1 to 0.05:1. Results from 1D displays of the ^{15}N HSQC studies revealed that the PAR3G$_{FD}$ D54 peak became more broadened than the F47 peak in the presence of ProT and PPACK-thrombin (Figure S4). Further information about interactions occurring with these two proteins was obtained by examining the 2D HSQC titrations (Figure 5A,B). In the presence of ProT, PAR3G$_{FD}$ residue D54 exhibited changes in chemical shift that resemble a fast exchange scenario corresponding to an interaction with weakened affinity. This interaction creates a new protein environment for the ligand. By contrast, PAR3G$_{FD}$ residue F47 residue exhibited little change in chemical shift during the ProT titration, suggesting this residue did not experience a substantial change in its chemical environment relative to that of the free peptide (Figure 5A). The project then proceeded to titrations with PPACK-thrombin where additional new effects were observed. The 2D HSQC crosspeak for D54 underwent extensive line broadening and could be observed for only thrombin:PAR3G$_{FD}$ ratios of 0.16:1 to 0.05:1 (Figure 5B). Such a peak broadening reflects improved interactions with the mature ABE I site and is consistent with intermediate exchange conditions. Interestingly, the F47 residue was observed to undergo a wider range of chemical shift changes for the 2:1 to 0.05:1 titration series. F47 was now experiencing a chemical environment different from what had been observed with ProT containing an immature (pro)-ABE I.

The chemical shift changes for the D54 and F47 titrations proceed in the same directions across the NMR panels when probing ProT versus PPACK-IIa (upfield for F47 and downfield for D54). These similarities are consistent with the PAR3 and PAR3G peptides interacting within the same binding region of ProT versus PPACK-IIa. Nonspecific binding to a separate area on ProT is not evident. In further support of this proposal, Andersen et al.[30] confirmed that hirudin (54−65) can bind specifically to the (pro)-exosites of prothrombin and thrombin. Moreover, HDX-MS studies demonstrated that hirudin and PAR3 (44−56) both bind to thrombin ABE I region 65−84.[19]

NMR HSQC titrations can be used to quantitatively characterize binding interactions for individual ^{15}N-labeled amino acid residues. These calculations work the best when there are distinct changes in chemical shift position when ^{15}N-labeled peptide amino acid residues interact with a target protein. Furthermore, the binding curves should approach saturation.[47,48] With the current project, the peptide concentration remained constant, and the protein solutions were serially diluted. For these protein−peptide ligand systems, affinity estimates could be made for some of the ^{15}N-labeled PAR3G (44−56) amino acid residues. On the basis of the titration data that were collected, D54 had a binding affinity

DOI: 10.1021/acs.biochem.7b00970
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Table 1. K_D and $|\Delta\omega_{max}|$ Values Determined from 2D HSQC Titrations for ^{15}N-Labeled PAR3G$_{FD}$ and PAR3G$_{EL}$ Bound to Human Prothrombin and Thrombin (wild type and mutant)[a]

peptide	residue	[ProT] (μM)	$\lvert\Delta\omega_{max}\rvert$ (ppm)	wild-type PPACK-IIa	$\lvert\Delta\omega_{max}\rvert$ (ppm)	R77aA PPACK-IIa
PAR3G$_{FD}$	F47	64 ± 8	0.23 ± 0.01	40 ± 10 μM	1.98 ± 0.2	173 ± 85 μM
PAR3G$_{FD}$	D54	65 ± 12	0.39 ± 0.03	too tight to calculate	insufficient data points	168 ± 88 μM
PAR3G$_{EL}$	L52	124 ± 27	0.37 ± 0.04	47 ± 6 μM	1.84 ± 0.08	salt bridge not involved
PAR3G$_{EL}$	E48	>200	NA	too tight to calculate	insufficient data points	salt bridge not involved

[a]For these NMR titrations, the peptide ligand concentrations were kept constant and the protein concentrations were serially diluted. Estimated K_D values were calculated using in-house scripts written in Python. Experimental data employed in the calculations include the individual protein and peptide concentrations and also the ^{15}N NMR chemical shift differences between each set of free and bound conditions. The plasma-derived ProT and IIa titration series were performed at least in triplicate. The studies with recombinant R77aA-IIa were performed in duplicate. Error analysis was performed using a Monte Carlo approach assuming a 10% error in the serially diluted protein samples. See Materials and Methods for more details.

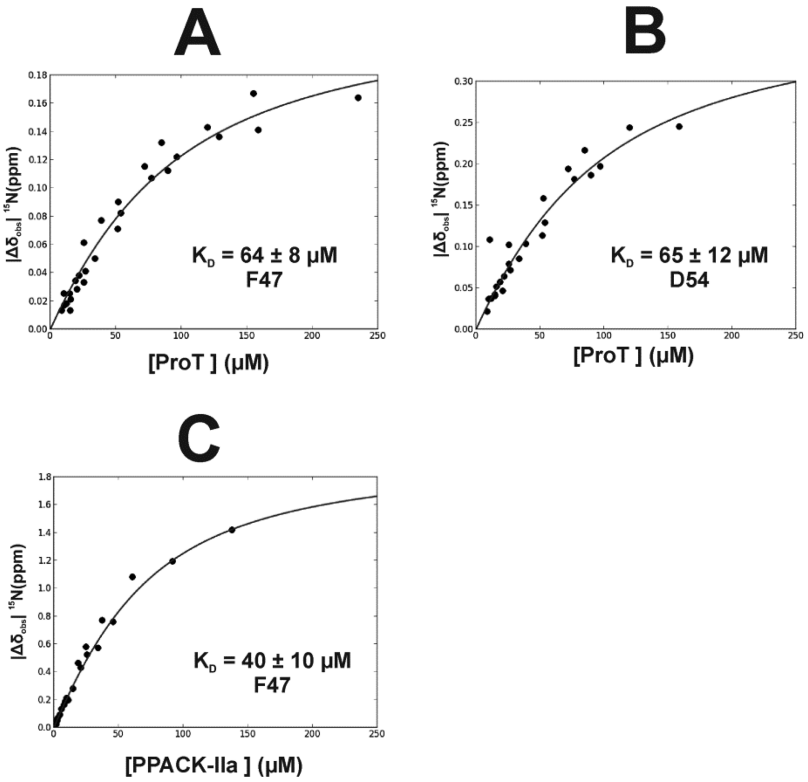


Figure 6. Determination of the binding affinity (K_D) for ^{15}N-labeled F47 and D54 of PAR3G$_{FD}$ interacting with prothrombin and PPACK-IIa. For this NMR titration series, the peptide ligand concentration was kept constant and the ProT and PPACK-IIa concentrations were serially diluted. As a result, the NMR titrations were measuring the binding of protein to a defined peptide concentration. Interactions between (A) ProT and PAR3G [^{15}N]F47 led to a K_D of 64 ± 8 μM, (B) ProT and PAR3G [^{15}N]D54 led to a K_D of 65 ± 12 μM, and (C) PPACK-IIa and PAR3G [^{15}N]F47 led to a K_D of 40 ± 10 μM. NMR titrations were performed in triplicate. The reported K_D values were determined using in-house scripts written using Python. The term $|\Delta\delta_{obs}|$ ^{15}N ppm = $\delta^{15}N_{bound} - \delta^{15}N_{free}$ reflects the absolute difference in chemical shift between the bound and free states of the particular ^{15}N residue. Error analysis was performed using a Monte Carlo approach assuming a 10% error in the serially diluted protein samples. See Materials and Methods for more details.

(K_D) of 65 ± 12 μM in the presence of ProT and a maximal chemical shift difference ($\Delta\omega_{max}$) of 0.39 ± 0.03 ppm (Table 1 and Figure 6B). The interaction became stronger with thrombin; however, the binding of D54 was too tight and the peak broadening too severe to determine a K_D value from the titration curve. The labeled PAR3 F47 residue had an estimated K_D value of 64 ± 8 μM and a $\Delta\omega_{max}$ of 0.23 ± 0.01 ppm for ProT and a K_D value of 40 ± 10 μM and a $\Delta\omega_{max}$ of 1.98 ± 0.19 ppm with thrombin (Table 1 and Figure 6A,C). Curiously, F47 experienced a marginal increase in affinity upon exosite maturation even though this residue clearly encountered a new environment reflected by its now larger chemical shift spread ($\Delta\omega_{max}$) (Table 1 and Figure 6C).

^{15}N HSQC NMR Titration Studies with PAR3G (44−56) Labeled at D54 and F47 Interacting with Thrombin Mutant R77aA. The HSQC titrations described above revealed a significant increase in affinity for PAR3 residue D54 as ProT is converted to thrombin. Acidic PAR3 D54 has been reported to make salt bridge contact with basic thrombin residue R77a[9] (Figure 2). Interactions between these two residues likely become more effective once the mature ABE I is formed. To probe whether electrostatic interactions are occurring between R77a and D54, ^{1}H−^{15}N HSQC titrations were performed with a thrombin R77aA−PAR3G$_{FD}$ complex (Figure 7). Unlike the wild-type PPACK-thrombin titrations, D54 and F47 chemical shifts could be followed for the full

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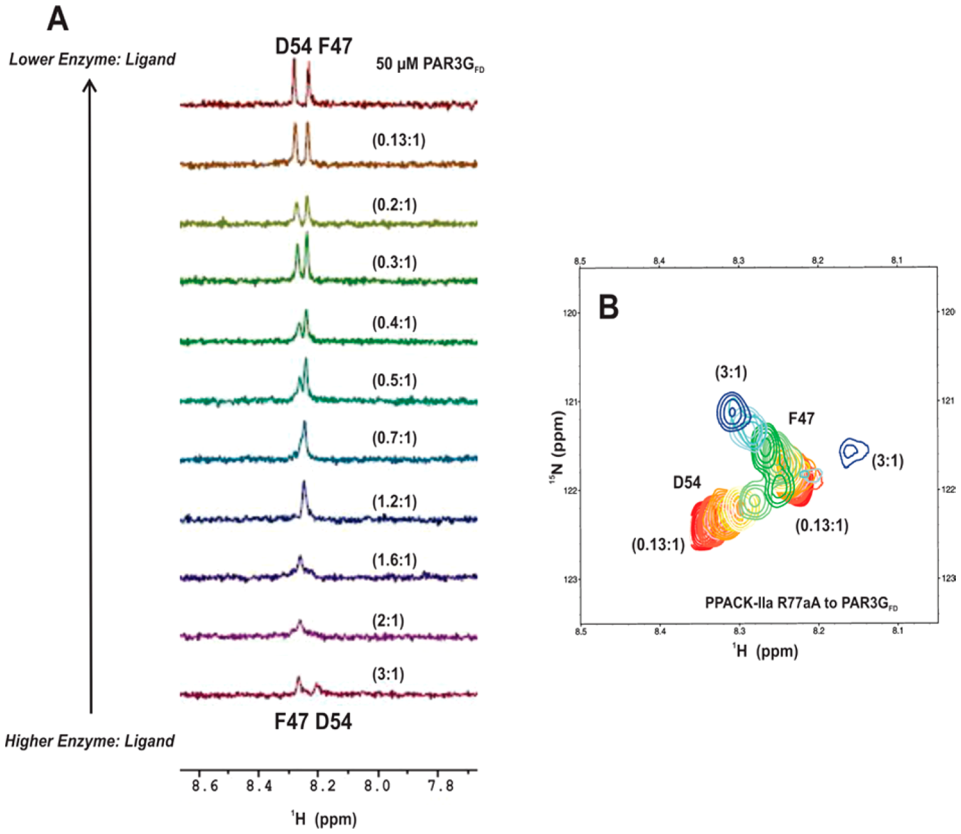


Figure 7. 1D and 2D ^{1}H−^{15}N HSQC NMR titrations of PAR3G$_{FD}$ (44−56) in the presence of PPACK-R77aA. All NMR samples were in 25 mM H_3PO_4, 150 mM NaCl, 0.2 mM EDTA, and 10% D_2O (pH 6.5). (A) For the 1D HSQC NMR titrations, the starting complexes included 50 μM PAR3G$_{FD}$ (44−56, [^{15}N]F47, [^{15}N]D54) in 150 μM PPACK-R77aA. The serial dilutions resulted in PPACK-R77aA:PAR3G$_{FD}$ ratios that spanned from 3:1 to 0.1:1. Note that at the start of the titration (3:1 protein:peptide ratio) the amide proton for PAR3G F47 was at 8.3 ppm and that for PAR3G D54 at 8.2 ppm. By a ratio of 0.7:1, the two peaks overlapped into a single peak. As the PPACK-R77aA thrombin was further diluted, the F47 and D54 peaks continued to change resonance positions and eventually matched those of the free PAR3G$_{FD}$ peptide. Representative data sets are shown. (B) For the 2D ^{1}H−^{15}N HSQC titrations, starting complexes included 50 μM PAR3G$_{FD}$ (44−56, [^{15}N]F47, [^{15}N]D54) in 150 μM PPACK-R77aA. The serial dilutions resulted in PPACK-thrombin:PAR3G$_{FD}$ ratios that spanned from 3:1 to 0.1:1. Representative data sets are shown. Colors for the 2D ^{1}H−^{15}N HSQC crosspeaks span from blue (highest protein:peptide ratio) to red (free peptide).

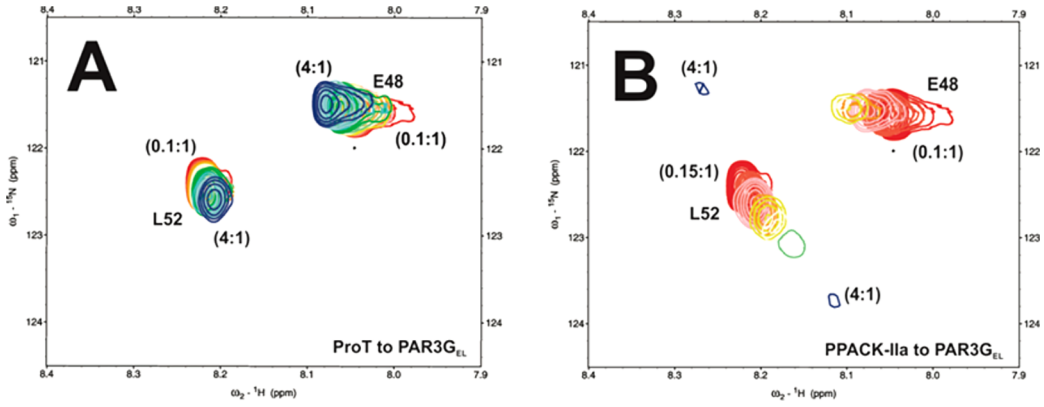


Figure 8. 2D ^{1}H−^{15}N HSQC NMR titrations of PAR3G$_{EL}$ (44−56) in the presence of ProT and PPACK-IIa. All NMR samples were in 25 mM H_3PO_4, 150 mM NaCl, 0.2 mM EDTA, and 10% D_2O (pH 6.5). (A) For the PAR3G$_{EL}$ binding studies with ProT, starting complexes included 50 μM PAR3G$_{EL}$ (44−56, [^{15}N]E48, [^{15}N]L52) in 180 μM ProT. The serial dilutions resulted in ProT:PAR3G ratios that spanned from 4:1 to 0.1:1. (B) For PPACK-IIa, starting complexes included 50 μM PAR3G$_{EL}$ (44−56, [^{15}N]E48, [^{15}N]L52) in 211 μM PPACK-IIa. The serial dilutions resulted in PPACK-IIa:PAR3G$_{EL}$ ratios that spanned from 4:1 to 0.1:1. Representative data sets are shown. Colors for the 2D ^{1}H−^{15}N HSQC crosspeaks span from blue (highest protein:peptide ratio) to red (free peptide).

titration series for PPACK-thrombin R77aA:PAR3G$_{FD}$ ratios of 3:1 to 0.13:1. The D54 crosspeak exhibited a broad range of chemical shifts and even overlapped with the F47 crosspeak for a portion of the titration (Figure 7A,B). Note in Figure 7A that at the start of the titration (3:1 protein:peptide ratio) the amide proton for PAR3G F47 was at 8.3 ppm and that for PAR3G D54 at 8.2 ppm. By a ratio of 0.7:1, the two peaks overlapped into a single peak. As the PPACK-R77aA was further diluted,

DOI: 10.1021/acs.biochem.7b00970
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the F47 and D54 peaks continued to change resonance positions and eventually matched those of the free $PAR3G_{FD}$ peptide. A similar set of trends can be observed in the 2D HSQC crosspeaks shown in Figure 7B. Overall, the 2D crosspeak patterns for both F47 and D54 were consistent with fast exchange conditions. For PAR3G D54, the estimated K_D was 168 ± 88 μM and $\Delta\omega_{max}$ equalled 1.83 ± 0.62 ppm, whereas for PAR3G F47, the values were a K_D of 173 ± 85 μM and a $\Delta\omega_{max}$ of 1.79 ± 1.05 ppm (Table 1 and Figure S5A,B). The removal of the salt bridge between PAR3 D54 and thrombin R77a clearly weakened the interaction of D54 with the ABE I surface, and a K_D value could now be estimated. In addition, an ∼3-fold loss of affinity was observed for PAR3G F47 in the presence of the R77aA thrombin mutant. Even with the weakened affinity, thrombin R77aA still incurred large structural changes as reflected in the >1.8 ppm changes in the maximal chemical shift.

^{15}N HSQC NMR Titration Studies with PAR3G (44−56) Labeled at L52 and E48. Two additional PAR3 residues were chosen to probe the environments of the 30s and 70s loop regions of (pro)-ABE I. E48 of PAR3 (44−56) makes a salt bridge with R75 of thrombin, and L52 of PAR3 (44−56) is positioned within a hydrophobic pocket containing thrombin residues F34, L65, and I82 (Figure 2).[9] As observed with F47 and D54, the two new PAR3 residues E48 and L52 also exhibited 1D proton line broadening when complexed with ProT and PPACK-IIa (Figure 4). Thus, HSQC NMR titrations could be performed with PAR3G (44−56) labeled at [^{15}N]E48 and [^{15}N]L52.

The titration ratios for ProT−$PAR3G_{EL}$ and PPACK-thrombin−$PAR3G_{EL}$ spanned from 4:1 to 0.1:1. For the ProT-bound complex, 1D and 2D displays of HSQC titration peaks could be followed for all the titrations points (Figure S6A and Figure 8). Unlike those of $PAR3G_{FD}$, the binding affinities for L52 and E48 were weaker, resulting in K_D values of 124 ± 27 and >200 μM, respectively (Table 1 and Figure S7A,B). Moreover, both E48 and L52 exhibited fewer changes in chemical shift position. These results suggested that E48 and L52 did not encounter much of a change in binding environment when they were tethered weakly to the surface of ProT. Furthermore, we propose that ProT R75 is less available for interacting with E48 of PAR3 than the distinct interactions already observed between ProT R77a and D54 of PAR3. When the peptide was titrated with PPACK-thrombin (Figure S6B and Figure 8B), L52 exhibited a greater change in chemical shift position. By contrast, E48 resembled intermediate exchange on/off the thrombin surface. Extensive peak broadening could already be observed at 1.2:1 protein:ligand ratios. In the presence of PPACK-thrombin, L52 showed a 3-fold improvement in K_D (47 ± 6 μM) and E48 underwent extensive peak broadening, preventing a K_D value from being calculated. For L52, a $\Delta\omega_{max}$ of 0.37 ± 0.04 ppm was obtained for ProT and increased to 1.84 ± 0.08 ppm upon conversion to thrombin (Table 1 and Figure S7A,C).

DISCUSSION

With thrombosis becoming a global disease burden, the urge to develop better anticoagulants has strengthened.[51] Novel oral anticoagulants[52,53] that target the active sites of FXa and thrombin[54] have shown much medical promise, but reversing their therapeutic activities during heavy bleeding scenarios can be challenging.[55,56] Drug candidates[27,28,57] that target thrombin ABE I are an alternative strategy. Thrombin is, however, proteolytically derived from the zymogen prothrombin (Figure 1). Unexpectedly, some ABE I ligands can already bind to the immature pro-ABE I site on this zymogen.[29−32,58] Future ABE I-directed therapeutics might therefore be designed to target or avoid specific regions of pro-ABE I on ProT versus ABE I on thrombin.

With our NMR titration approaches, we characterized, for the first time, the binding of ABE I-directed PAR3 peptides to ProT versus PPACK-thrombin. Previous NMR relaxation studies had already revealed that communication from ABE I toward the active site region is still preserved with PPACK-thrombin.[37] Our 1D ^{1}H line broadening NMR results demonstrated that PAR3 (44−56) and its weaker binding version, PAR3G (44−56), could interact with both ProT and PPACK-thrombin (Figure 4).[9] NMR ^{1}H−^{15}N HSQC titration studies were then performed with a series of ^{15}N-labeled PAR3 peptides to characterize our ability to bind ProT versus PPACK-thrombin (Figures 5−8 and Figures S1−S6). Such an approach has the distinct advantage of monitoring exosite maturation at the individual amino acid residue level. Isothermal titration calorimetry, fluorescence, and surface plasmon resonance all provide a global K_D fit using all peptide amino acid residues. NMR, by contrast, preserves the ability to measure individual binding interactions in solution and thus document their contributions. Moreover, NMR is well suited for protein-ligand systems in the fast exchange regime.

For the current NMR project, PAR3 peptides were prepared with [^{15}N]amide labeling of F47, E48, L52, and D54. HSQC titrations were performed at the typical NMR pH of 6.5. The acidic and hydrophobic residues [^{15}N]D54 and [^{15}N]F47 on PAR3 (44−56) were examined first. This native PAR3 sequence bound too tightly to thrombin, hindering our ability to assess K_D values by NMR titration methods. Prior fluorescence titrations revealed a K_D of 2 μM for a related PAR3 sequence, thus confirming our observation of intermediate/slow exchange conditions in the NMR titrations.[59] Greater success was achieved with $PAR3G_{FD}$ (44−56) in which P51 was replaced with G51. With ProT, PAR3G D54 and F47 both had individual K_D values in the range of 65 μM. The immature pro-ABE I can thus accommodate both residues to a similar extent. Upon maturation to ABE I, PAR3G D54 bound too tightly to PPACK-thrombin for binding affinity calculation. By contrast, the F47 binding affinity increased only modestly as ProT was converted to thrombin. Interestingly, the F47 also documented a substantial change in the chemical environment ($\Delta\omega_{max}$ = 1.8 ppm) within the 30s loop region following activation to thrombin. Structural rearrangements are proposed to occur in the vicinity of ProT/thrombin residue F34 but do not result in a significant change in affinity (Figure 2).

The PAR3G D54 residue was hypothesized to be influenced by its electrostatic interaction with thrombin R77a (Figure 2).[9] This acidic PAR3 residue may have properties similar to those of hirudin.[60] The hirudin sequence (^{56}FEEI^{59}) has already been proposed to electrostatically steer toward the ABE I surface.[60,61] Ionic tethering between acidic hirudin residues and specific basic residues on ABE I might then be promoted followed by stabilizing hydrophobic interactions.[60,61] These hypotheses could now be tested with the PAR3G sequence and thrombin R77aA. As predicted, the affinity for D54 decreased 3-fold upon loss of its salt bridge partner. In response, F47 also displayed a much lower affinity value.

PAR3G residues E48 and L52 provided the opportunity to probe two additional regions of ProT versus PPACK-thrombin.

DOI: 10.1021/acs.biochem.7b00970
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E48 makes a salt bridge with thrombin R75, a neighboring region of the 70s loop. PAR3G L52 interacts with a cluster of hydrophobic residues, including thrombin F34, L65, and I82 (Figure 2).[9] Curiously, both E48 and L52 exhibited only minor changes in chemical shift upon binding to ProT. Furthermore, the estimated individual K_D values for L52 and E48 bound to ProT were ≥2-fold weaker than those for D54 and F47. The pro-ABE I surface on ProT may not accommodate these two residues as well as it does PAR3 D54 and F47. Similar to PAR3G D54, there was extensive line broadening for E48 upon conversion to thrombin, and K_D values could not be determined. By contrast, L52 exhibited a broad series of chemical shift changes during the thrombin titration ($\Delta\omega_{max}$ = 1.8 ppm). Interestingly, the estimated K_D for binding of L52 (K_D = 47 ± 6 μM) to PPACK-thrombin was similar to the value determined for F47 (K_D = 40 ± 10 μM). Both L52 and F47 interact at the more hydrophobic surface area of ABE I with a common overlapping influence from thrombin F34 (Figure 2).

CONCLUSIONS

Overall, our solution NMR studies have elucidated changes in the exosite environment that occur as zymogen ProT is converted to active thrombin. 1D proton line broadening NMR and $^{1}H-^{15}N$ HSQC studies demonstrated that PAR3G E48 and D54 could already interact with ProT and the affinity increased upon maturation to thrombin. In the NMR studies, the 3-fold tighter binding affinity for D54 versus that of E48 suggests that the 70s loop region surrounding thrombin R77a may be better oriented to bind PAR3G D54 than the interaction between thrombin R75 and PAR3G E48 (Figure 2). Moreover, a close review of the crystal structure overlays in Figure 3 suggests that the 70s loop region of ProT may assume an orientation that is preconfigured to that of PAR3-thrombin and PPACK-thrombin. As the ionic PAR3G D54–ProT/IIa R77a and PAR3G E48–ProT/IIa R75 interactions are stabilized, PAR3G F47 and L52 are proposed to take further advantage of interacting with thrombin F34 (within the 30–40s loop) and the thrombin hydrophobic cluster of F34, L65, and I82 (Figure 2). Aromatic PAR3G F47 and aliphatic PAR3G L52 both reported on significant changes in the chemical environment upon conversion of ProT to thrombin. The region surrounding the ProT/IIa 30s loop appears to be more affected than the hydrophobic pocket containing F34, L65, and I82.

Curiously, no striking differences are observed between the crystal structures of immature pro-ABE I in ProT versus the mature ABE I regions of thrombin–ligand complexes (Figure 3). Moreover, there are no X-ray crystal structures for a ligand bound to the pro-ABE I region of ProT. We can speculate that there are transient structural states within this coagulation protein system that are differentially explored by ProT and thrombin. Some of these states may be shared, whereas others may be unique. Interestingly, ProT already possesses a binding competent state. Additional research has implicated transient, or lowly populated, structures as playing an essential role in molecular interactions.[62−64] Thus, we believe that the mature thrombin likely visits a binding competent state more often than ProT does, thereby promoting the higher affinity of PAR3 for thrombin.

In conclusion, our NMR titration studies have clearly revealed that individual PAR3 amino acid residues are documenting structural rearrangements that occur upon ABE I maturation. Moreover, these amino acid residues make individual contributions to the overall binding affinity. The knowledge gained from this NMR project may be used to help decipher the characteristic features of pro-ABE I versus ABE I that become accessible to physiological ligands or future drug candidates.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.biochem.7b00970.

2D $^{1}H-^{13}C$ natural abundance HSQC NMR spectra of PAR3 (44–56) and PAR3G (44–56) (Figure S1), 1D ^{15}N HSQC NMR titrations of $PAR3_{FD}$ (44–56) in the presence of ProT and PPACK-IIa (Figure S2), 2D ^{15}N HSQC NMR titrations of $PAR3_{FD}$ (44–56) in the presence of ProT and PPACK-IIa (Figure S3), 1D ^{15}N HSQC NMR titrations of $PAR3G_{FD}$ (44–56) in the presence of ProT and PPACK-IIa (Figure S4), 1D ^{15}N HSQC NMR titrations of $PAR3G_{EL}$ (44–56) in the presence of ProT and PPACK-IIa (Figure S5), determination of the binding affinity (K_D) for ^{15}N-labeled F47 and D54 of $PAR3G_{FD}$ interacting with PPACK-R77aA IIa (Figure S6), and determination of the binding affinity (K_D) for ^{15}N-labeled L52 and E48 of $PAR3G_{EL}$ interacting with prothrombin and PPACK-IIa (Figure S7) (PDF)

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R.B., T.M.S., and M.C.M. designed the NMR research. R.B. performed the NMR experiments. R.B. and D.B. analyzed the NMR data. R.B., D.B., T.M.S., and M.C.M. critically evaluated the results and wrote the manuscript. M.C.M. supervised the research.

Funding

This research was supported by National Institutes of Health Grants R01HL68440 (M.C.M.) and R15HL120068 (M.C.M.), a University of Louisville Research Initiation Grant (M.C.M.), and by start-up funds provided by the James Graham Brown Foundation (T.M.S.).

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

The authors thank N. Stolowich for help with NMR instrumentation. The authors are grateful for the generous gift of thrombin mutant R77aA provided by Enrico Di Cera and Leslie Pelc from the St. Louis University School of Medicine. The authors thank M. V. Malovichko for initiating HSQC titration projects with thrombin ligands in the lab. Finally, the authors appreciate critical discussions on this research with K. N. Mouapi and B. A. Anokhin.

DOI: 10.1021/acs.biochem.7b00970
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DOI: 10.1021/acs.biochem.7b00970
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